

7824 Zero Road, second floor

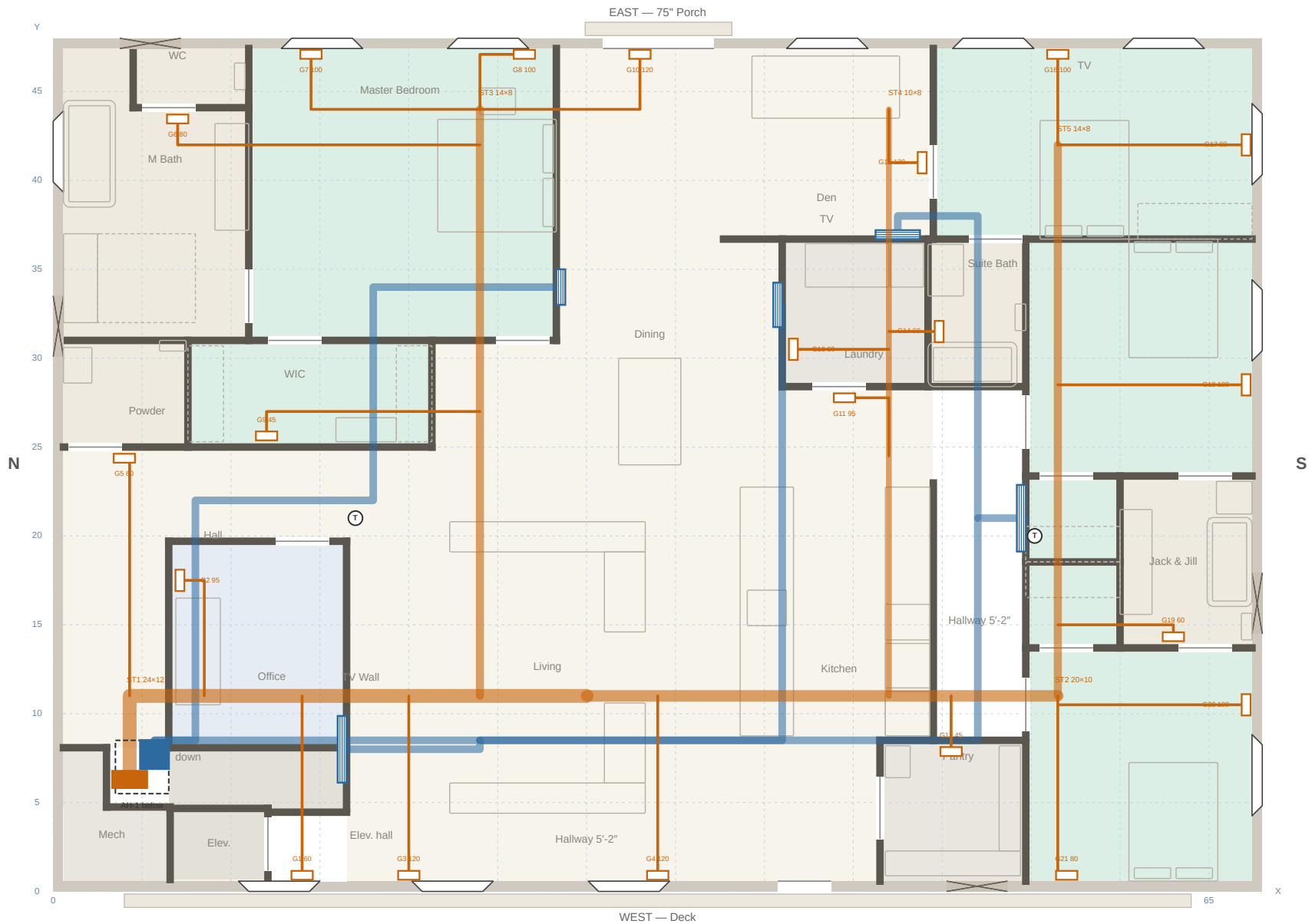
M-6 HVAC plan · packet for the hvac contractor

Supply and return ductwork in the floor trusses from AH-1 below, floor registers and returns, zoning. Read it with the hydronic pack, sheets M-1 to M-5 (packets/M-1-to-M-5-hydronic-pack.pdf): the radiant floor heats this floor and shares the truss bays with these ducts.

Rev C · 2026-10-01 · Schematic for pricing and coordination, not a stamped construction document. Verify every dimension on site. Coordinates are feet from the north and west brick faces; the plan is drawn with east at the top and north on the left. Wall numbers match the concept drawing. Questions: name the sheet and the tag (for example M-6 G7).

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7824 Zero Road — second floor

Casper, Wyoming · 68' x 48' to brick, faces 7" inside · 9' ceiling · grid 5'
Wall numbers as the floor plan drawing of 2026-09-29

HVAC plan

Supply trunks, branches and floor registers in the trusses; returns; AH-1 below
Schematic for pricing and coordination, not a stamped construction document. Verify on site

M-6

rev C · 2026-10-01
drawn for K. Christianson

Notes

1. AH-1, the 6-ton hydronic air handler (2,400 CFM) on hydronic sheet M-3, sits on the floor below at X 5', Y 7', directly under the stair landing. It serves both floors through the D-UP and D-DN dampers on sheet M-5; this sheet ducts the second floor only. The first floor gets its own trunk off the handler later and is not routed through this floor. Supply riser 24" × 12" and return riser 20" × 20" come up through the floor into the 24" open-web truss space.
2. Airflow split, from the thermal model (owner, 2026-10-01): each floor's trunk is sized for its own job, not a share of 2,400. This floor is heated by the radiant floor (about 95% of its heat; the air handler heats it only about 60 hours a year), so its ducts are sized for cooling: about 2.5 tons on the design day, held at 68°F, which takes 1,300 to 1,650 CFM on 45°F chilled water. Design airflow here is 1,800 CFM (D-UP 24" × 12"); the floor below is sized for heating at about 1,200 CFM (D-DN 16" × 12") and never calls for cooling in the model. The two add up to 3,000, more than the fan's 2,400, which is fine: both floors need air at once only about 50 hours a year, and then the dampers share it. Roughly 60% of the design air goes up and 40% down.
3. Heat on this floor is the radiant floor on hydronic sheet M-1: PEX in every truss bay, zones A, B and C on thermostats T1 to T3. AH-1 heats the floor below (hydronic sheet M-4) and runs heat-only until the Phase 2 heat pump HP-1 brings cooling.
4. When only this floor calls, run the ECM blower at about 1,800 CFM; until the first floor is ducted that is the handler's top speed. The ducts here are sized for 1,800 CFM and would be noisy and over-pressured at the full 2,400, so cap the blower at about 1,800 with D-DN closed (or fit a barometric bypass at the handler).
5. Every duct runs inside the floor trusses: rectangular trunks ST1 to ST5 at the truss bottom chord, 5", 6" and 7" round branches to floor registers set tight to the walls, mostly the exterior walls under the windows: the near edge of each register sits 2" off the finished wall face, long side along the wall. Trunks and branches are sized at 0.08" per 100' friction with trunks at 900 fpm or less: ST1 24" × 12" leaves the riser with the full 1,800 CFM, ST2 20" × 10" carries the 1,020 CFM left east of X 30'. Registers total 1,800 CFM; the room split follows the earlier first-pass sizing scaled up, and the HVAC contractor balances the rooms with a Manual J / D.
6. No floor grilles anywhere (wheelchair). Returns are wall grilles at the baseboard, 4" to 12" AFF: R1 on the living face of wall 27, R2 on the bedroom face of wall 19, R3 on the hall face of wall 7, R4 on the den face of wall 17, R5 on the dining face of wall 32. A lined stud bay passes about 250 CFM in a 2x6-plate wall and 180 in a 2x4 wall, so each return spans the number of bays listed (one 14×8 grille per bay, side by side) and the five together pass the full 1,800 CFM and the bays join in a plenum box in the truss space before the rectangular duct runs back to the return riser. Walls 17, 19, 27 and 32 are 2x6-plate staggered walls; wall 7 is 2x4.
7. Supply registers are floor registers because the air comes up through the floor; they are flush, heavy-gauge steel, screwed to the floor, 4" wide with no raised lip, so a wheel crosses them like the floor. If even that is unwelcome, say so: the alternative is toe-kick registers under the island and vanities plus baseboard supplies in the 2x6 walls, at the cost of less even heat under the windows.
8. Two air zones with motorised dampers at the branch trunks: zone 1 public rooms and the master suite, zone 2 the bedroom wing. Thermostats T6 on the office wall facing the hall and T7 in the wing hall, 48" AFF (T1 to T5 are the hydronic pack's, sheet M-5). See the zoning open item.
9. Bath exhaust, the dryer and the cooktop downdraft are on P-1 (exhaust for coordination). The ERV in the mech room ducts fresh air into the return trunk at the riser and exhausts from the baths; its 6" ducts also run in the floor.
10. Floor registers: steel 4" × 10" / 12" / 14" with opposed-blade dampers, flush, no raised lips (wheelchair). Keep registers 12" clear of pocket-door cavities and the drains on P-1; the drain mains at Y 21'-6" and X 40' cross the trunks and win the lower position.
11. Insulate supply trunks in the truss space (R-6 wrap); seal every joint with mastic; test to 4 CFM per 100 sq ft.

Open items

- Equipment: AH-1 serves both floors from its hydronic coil (B-1 now, HP-1 later; hydronic sheet M-3). Confirm its external static (0.5" assumed) and that its ECM blower can hold 1,800 CFM to this floor. MBTEK coil data is still needed to confirm the 1,800: sensible cooling at 45°F and 50°F water and heating at 120, 130 and 140°F, each at about 1,000 and 1,800 CFM. On 50°F water this floor would need 1,750 to 2,400 CFM to hold 68°F, so plan on 45°F chilled water. Floor-to-floor zoning is the D-UP and D-DN damper pair on hydronic sheet M-5.
- Zoning: the hydronic pack runs the air side from T4 downstairs, with one damper (D-UP) for this floor (sheet M-5). Keep the two air zones drawn here, which adds T6, T7 and a zone panel to the M-5 schedule, or run this floor as one zone on D-UP with balancing dampers only?
- Radiant floor (hydronic sheet M-1): PEX runs down and back in every truss bay 2" below the subfloor, R-19 batts under it, and the loop leaders run along the west wall through truss web holes. Every register boot and branch here passes through that layer, and the five west-wall registers (G1, G3, G4, G12, G21) land where the leaders and loop ends run. Lay the registers out between PEX passes before the tubing goes in, and give the truss maker the leader holes with the duct routes.
- Truss coordination: ST1 (24" × 12") and ST2 (20" × 10") run along Y 11' across the trusses, so the truss maker needs a duct chase about 26" wide and 14" high in every truss from X 4' to X 57', below the radiant batts. ST2 and the 4" drain M3 cross at X 40'; ST3 (14" × 8") and the 3" drain M1 cross at Y 21'-6"; the truss manufacturer needs all of these routes.
- Whether the deck-side (west) registers can sit under the windows as drawn or should move to the toe-kick of the island and sofas.

Schedules

Registers and returns

Tag	Room	Position	Grille	CFM	Duct	Run
G1	Elevator hall	X 14'-0", Y 11"	4×10	60	5"	10'-1"
G2	Office	X 7'-2", Y 17'-6"	4×12	95	6"	7'-10"
G3	Living, west	X 20'-0", Y 11"	4×14	120	7"	10'-1"
G4	Living, west	X 34'-0", Y 11"	4×14	120	7"	10'-1"
G5	Powder / hall	X 4'-0", Y 24'-4"	4×10	60	5"	13'-8"
G6	Master bath	X 7'-0", Y 43'-6"	4×12	80	6"	18'-6"
G7	Master bedroom	X 14'-6", Y 47'-1"	4×12	100	6"	12'-7"
G8	Master bedroom	X 26'-6", Y 47'-1"	4×12	100	6"	5'-7"
G9	Walk-in	X 12'-0", Y 25'-8"	4×10	45	5"	13'-4"
G10	Dining / east entry	X 33'-0", Y 47'-1"	4×14	120	7"	12'-1"
G11	Kitchen	X 44'-6", Y 27'-9"	4×12	95	6"	5'-9"
G12	Pantry	X 50'-6", Y 7'-10"	4×10	45	5"	3'-2"
G13	Laundry	X 41'-8", Y 30'-6"	4×10	60	5"	5'-4"

Tag	Room	Position	Grille	CFM	Duct	Run
G14	Suite bath	X 49'-10", Y 31'-6"	4×10	60	5"	2'-10"
G15	Den	X 48'-10", Y 41'-0"	4×14	120	7"	4'-10"
G16	2nd Suite	X 56'-6", Y 47'-1"	4×12	100	6"	5'-1"
G17	2nd Suite	X 67'-1", Y 42'-0"	4×12	80	6"	10'-7"
G18	Bedroom 4	X 67'-1", Y 28'-6"	4×12	100	6"	10'-7"
G19	Jack & Jill	X 63'-0", Y 14'-4"	4×10	60	5"	7'-2"
G20	Bedroom 3	X 67'-1", Y 10'-6"	4×12	100	6"	10'-7"
G21	Bedroom 3	X 57'-0", Y 11"	4×12	80	6"	10'-7"
R1	Living, wall 27	X 16'-6", Y 8'-0"	3 bays, 3 × 14×8 wall	530	14×8	27'-1"
R2	Master bedroom, wall 19	X 28'-4", Y 34'-0"	1 bay, 1 × 14×8 wall	250	8×8	48'-1"
R3	Wing hall, wall 7	X 54'-8", Y 21'-0"	3 bays, 3 × 14×8 wall	420	12×8	43'-2"
R4	Den, wall 17	X 47'-6", Y 36'-8"	2 bays, 2 × 14×8 wall	300	10×8	22'-10"
R5	Dining, wall 32	X 41'-0", Y 33'-0"	2 bays, 2 × 14×8 wall	300	10×8	41'-6"

Supply trunks

Trunk	Size	Length	Route
ST1	24×12	30'-5"	main trunk, 1,800 CFM (this floor's cooling airflow), east under the passage, office and living
ST2	20×10	26'-6"	1,020 CFM on, under the kitchen and the wing hall to the bedrooms
ST3	14×8	33'-0"	445 CFM north to the master suite and the east entry
ST4	10×8	33'-0"	335 CFM to the kitchen, laundry, suite bath and den
ST5	14×8	31'-0"	520 CFM under Bedrooms 3 and 4 and the 2nd Suite

Bill of materials

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The contractor takes off the final list.

Item	Qty	Notes
Rectangular supply trunk 24×12, galvanised, R-6 wrapped	30 lf	plus 10% for fittings
Rectangular supply trunk 20×10, galvanised, R-6 wrapped	26 lf	plus 10% for fittings
Rectangular supply trunk 14×8, galvanised, R-6 wrapped	64 lf	plus 10% for fittings

Item	Qty	Notes
Rectangular supply trunk 10×8, galvanised, R-6 wrapped	33 lf	plus 10% for fittings
Rectangular return duct 14×8, galvanised	27 lf	
Rectangular return duct 12×8, galvanised	43 lf	
Rectangular return duct 10×8, galvanised	64 lf	
Rectangular return duct 8×8, galvanised	48 lf	
Round branch 5", insulated flex or snap-lock	77 lf	7 runs, 3' each extra for the boot
Round branch 6", insulated flex or snap-lock	128 lf	10 runs, 3' each extra for the boot
Round branch 7", insulated flex or snap-lock	49 lf	4 runs, 3' each extra for the boot
Floor register 4×10 with damper, flush steel	7	with floor boot
Floor register 4×12 with damper, flush steel	10	with floor boot
Floor register 4×14 with damper, flush steel	4	with floor boot
Wall return grilles 14×8 at the baseboard, one per stud bay	11	5 returns; lined bays joined by plenum boxes in the truss space
Take-off collars with dampers	21	one per branch
Zone dampers, motorised, with a 2-zone control panel	2	on ST3/ST4 and ST5; see the zoning open item
Thermostats	2	T6, T7, 48" AFF; see the zoning open item
Supply riser boot 24×12 and return riser boot 20×20 through the floor	1 each	at X 4'-4"/Y 6'-4" and X 5'-8"/Y 7'-8"; coordinate the floor opening with the truss layout
Mastic, tape, hangers	lot	

Quantities are rough-order take-offs from the drawing geometry, for pricing. The contractor takes off the final list.